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Evaluation the DASS-8 as a rapid screening tool for mental health in the Malaysian population

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Abstract

Background Rising rates in depression and suicide across the globe highlight the urgent need for reliable screening tools to identify psychological distress. The 8-item Depression, Anxiety, Stress Scale (DASS-8) offers potential as a brief, efficient measure of psychological wellbeing, but requires validation in Malaysian populations. This study aimed to conduct comprehensive psychometric testing of the DASS-8 in a Malaysian sample.

Methods Study 1 involved 665 Malaysian participants who completed an online survey. Exploratory factor analysis (EFA) was performed to examine the internal structure of DASS-8. Study 2 included 202 participants who completed surveys before and after an experimental task to assess short-term temporal stability. Confirmatory factor analysis (CFA) and reliability analysis were conducted.

Results The DASS-8 demonstrated excellent psychometric properties with high internal consistency (Cronbach's $\alpha = 0.86$). Both EFA and CFA revealed a unidimensional factor structure, contrasting with the three-factor structure (depression, anxiety, stress) found in longer DASS versions. Short-term temporal stability and interclass correlation coefficients supported the scale's temporal stability (ICC = 0.87, 95% CI [0.83, 0.90]).

Conclusion The DASS-8 shows strong psychometric properties in Malaysian samples and functions effectively as a brief measure of general psychological distress rather than as a tool for distinguishing between specific emotional states.

Keywords DASS-8, Psychometric evaluation, Psychological distress, Malaysian sample

1 Introduction

It is estimated that one in seven people across the globe population is suffering from acute mental distress, with mood disorders related to anxiety and depression being the most common conditions [1]. In total, researchers estimate that 20–30% of the world population will at some time in their lives grapple with a mental disorder [2, 3] with mood disorders accounting for five out of every six adult cases of mental illness [4]. Although the precise aetiology of mood disorders is not fully understood [5], it is generally accepted that anxiety and depressive disorders are major causes of suicide [6]. Therefore, it is concerning that current trends indicate an increase in levels of stress, anxiety and depression, particularly among young adults [5].



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Asia has the highest global suicide rates [7, 8], and in the Southeast Asian nation of Malaysia, approximately 25% of young adults exhibit symptoms of elevated stress, anxiety, and depression [9]. The potential costs of increases in conditions associated with mood disorders are significant in terms of their direct impact on healthcare costs and lost productivity [10] through to the highest personal cost of suicide [11, 12]. Concerningly, the increases in the prevalence of stress, anxiety and depression in Malaysia has coincided with increases in the suicide rates which have risen from 641 in 2020 to 1068 in 2023 [13]. Efforts to mitigate this danger must be taken, including the formulation of appropriate preventative measures, screening and diagnostic tools.

Recent research suggests that anxiety and depression are maladaptive responses to stress [14]. Anxiety and depression are both associated with suicide, with levels of suicide being significantly higher among people suffering from these disorders [6, 15]. However, stress and anxiety alone are a relatively minor direct contributors to suicide [16], but anxiety when contributing to or combined with depression is associated with higher levels of suicide [17]. Approximately 50% of those exhibiting depressive symptoms also exhibit anxiety, but it appears that anxiety may emerge earlier in the lifespan than depression, and that anxiety may be a precursor to a large proportion of depressive disorders [18]. Thus, it appears that in a significant proportion of cases, stress may precede anxiety, which in turn may be an antecedent of depression. Indeed, depression develops across a spectrum of stages, beginning with subthreshold depression [19, 20]. Subthreshold depression is caused by gradually increasing levels of disturbance in the neurological underpinnings of the affective processes which are responsible for emotions and mood [21, 22]. Although subthreshold depression does not meet the diagnostic criteria for major depressive disorder [23], it remains a considerable public health burden due to its damaging impact on individuals' quality of life, personal economics, and overall cognitive functioning [24–26]. Subthreshold depression is more prevalent than clinical depression [23] and is a significant risk factor for the eventual development of depression [20]. However, current approaches to diagnosing depressive mental illnesses lack consistency [27] and are often only applied after the underlying mood disorders have worsened over long periods of time into major depression [28]. Therefore, it is crucial to identify the onset and development of depressive disorders and poor psychological wellbeing at the earliest possible stage so that effective interventions can be made to prevent disturbances that may lead to self-harm or suicide.

An additional factor to consider in the development of depression is the extent to which cultural differences may influence affect, since the definition of “normal” affective functioning is highly dependent on a range of factors including culture, ethnicity, and religion [29]. For example, East-Asian people have been found to suppress or avoid expressing their emotions relative to Western people [30] and may also be less prone to develop mood disorders [31]. Specific cultural characteristics, such as power distance, may also be associated with the presentation of mental disorder [32]. As frameworks of affective processes develop to incorporate cultural influence [33, 34], so too should the mood disorder discourse adapt reflect possible cultural differences in aetiology or symptoms, such as variations of irrational ideation [35].

Indeed, cultural differences in mental health are already evident within Malaysia, where 57.5% of local suicide cases over the past 10 years involved Chinese Malaysians [36] despite this group only being 22.4% of the population [37]. Cultural beliefs can shape

attitudes towards mental health, and assumptions about the causes of mental health difficulties [38, 39], which can in turn play a role in moderating mental health [40].

In Malaysia, mental health difficulties can be a source of considerable stigma [41, 42], particularly in rural regions and among older populations [43]. Malaysians with a history of suicidal ideation have experienced various forms of stereotyping, prejudice, and discrimination [44], making it difficult for sufferers to discuss mental health challenges with friends, family [45, 46], and coworkers [47]. Stigma is so pervasive that even local news outlets have a tendency to generalise and misrepresent those with mental disorders, branding them as unpredictable, dangerous [48], or detrimental to society [49]. Stigma may help to exacerbate mental health conditions by discouraging help-seeking behaviour that may lead to diagnosis and treatment [45, 50, 51]. Therefore, attempts have been made to develop screening and therapeutic approaches such as remote teletherapy, which can afford patients greater convenience and privacy to avoid public stigma [52]. However, if such approaches are to be successful then they must be supported by screening tools that are efficient, effective and scalable [53, 54]. One such screening tool is the Depression, Anxiety, and Stress Scale [55] from which a shorter 21-item version (DASS-21) was later developed [56]. DASS-21 is a widely accepted screening tool that has been translated into at least 30 languages [57]. Its widespread use and the fact that it measures stress, anxiety, and depression suggests that the DASS-21 has the potential to be an effective screening tool in Malaysia that is capable of detecting the antecedents and onset of mood disorders.

However, despite the fact the DASS-21 has generally performed well across different populations [58, 59], recent research has raised questions about the stability of its conventional three-factor model in non-Western samples [60]. Recent studies [61–64] have demonstrated that alternative factor structures of the DASS are theoretically plausible: while a conventional three-factor model distinguishes between depression, anxiety and stress, a unidimensional model may reflect a general distress construct, and a bifactor model accommodates both a general construct and specific factors. For example, a bifactor model was found to best fit Jordanian [61] and Turkish [62] samples. These findings have led some to suggest replacing the three scales of the DASS-21 with a total score [63] as a general measure of psychological wellbeing or distress as opposed to screening tool for specific mood disorders [64–67]. Others suggest a shortened version of the scale could suffice for such purposes [68–70], and the eight-item Depression, Anxiety, and Stress Scale [71] was thus developed for situations in which a brief and general assessment is required. Brevity also has other benefits, including shortening the duration of clinical intake screenings that might otherwise last 45–90 min [72] and making it easier to gather information from those with impairments [73]. The availability of a short and valid mental health assessment would give practitioners greater flexibility to decide on the scope of intake data required depending on the privacy, complexity or urgency of the situation at hand [74]. Use of the DASS-8 among American, Australian, Ghanaian, and Saudi Arabian samples yielded three factors with satisfactory convergent, criterion, discriminant, and predictive validity coefficients, and evidence of measurement invariance and internal consistency [71, 75]. Given the mixed findings regarding the dimensionality of the DASS-21 and shorter versions such as the DASS-8 across different populations, it is important to evaluate the factor structure of the DASS-8 empirically to assess its potential as a brief, remotely administered screening tool for mood disorders

in Malaysia. Providing a robust assessment of mood symptoms online would eliminate the stigma that could be associated with the need to physically attend a clinic, and would also offer clinicians a ready means to assess the severity of any symptoms that might be precursors or indicators of depression. However, given the relative novelty of the DASS-8, to date no empirical research has been conducted to evaluate the psychometric characteristics of the questionnaire with Malaysian participants. Therefore, the present study addresses this gap in the literature by assessing the psychometric characteristics of DASS-8 in a general, non-clinical sample as a potential brief screening tool in the fight against rising suicide rates in Malaysia.

2 Study 1 methods

2.1 Participants and procedures

A total of 665 participants took part in Study 1, with 188 males (28.27%), 473 females (71.13%), and 4 others (0.60%). The inclusion criteria were that participants were Malaysian adults aged 18 or older, with no specific exclusion criteria. The participants were Malaysians above 18 years old ($M = 20.89$, $SD = 2.94$) who were proficient in English voluntarily agreed to complete an online survey after providing their informed consent. An opportunity sample of adult participants was recruited through social media and university information boards. The anonymous online survey via SurveyMonkey was completed in approximately 10–15 min with no compensation. The study was approved by the university's research ethics committee. Data collection for Study 1 took place between July and September 2022.

2.2 Measurements

2.2.1 DASS-8

The 8-item Depression, Anxiety and Stress Scale (DASS-8 [75]) is a shorter version of original DASS-42 and DASS-21 scales aimed at measuring depression, anxiety and stress. The DASS-8 was administered in English. Participants were asked to indicate how much the statement applied to them over the past week (e.g., I felt that I was using a lot of nervous energy) using a four-item Likert scale (1 = never; 4 = almost always). This scale has good internal consistency with a Cronbach's alpha of 0.86 in Study 1.

2.3 Statistical procedures

Descriptive analysis was performed using R 4.5.1 [76] using *tidyverse* package to examine mean, standard deviation, and data normality. Next, exploratory factor analysis (EFA) was performed using *psych* package to examine the internal structure of DASS-8. The number of factors retained was determined based on the Kaiser criterion (i.e., factors with eigenvalues > 1.0), as well as inspection of the scree plot and parallel analysis. Items with factor loadings ≥ 0.50 were considered salient and retained for interpretation [77]. Lastly, Cronbach's alpha was implemented to assess the internal consistency of DASS-8.

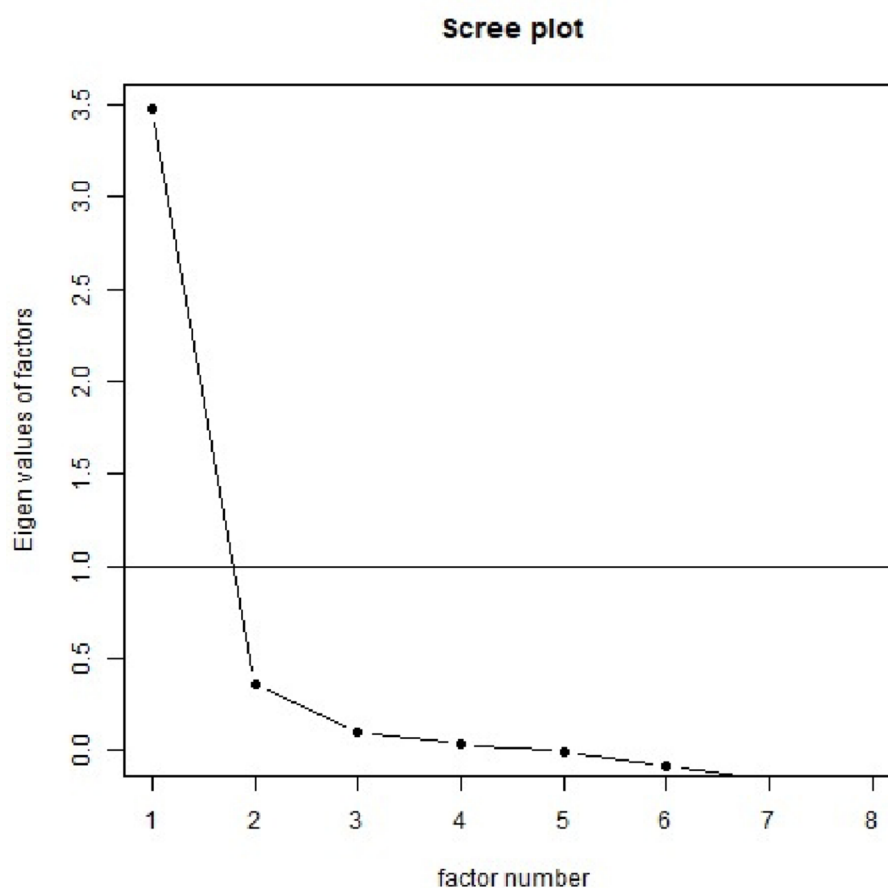
3 Study 1 results

3.1 Descriptive statistics

Table 1 presents the item-level descriptive statistics for DASS-8. It is noted that all items exhibited a normal distribution, as evidenced by skewness and kurtosis values falling within the acceptable range of ± 2 .

Table 1 Descriptive analysis for items in DASS-8

Item	Mean	SD	Skew	Kur
1. I felt that I was using a lot of nervous energy	2.37	0.71	0.58	0.11
2. I was worried about situations in which I might panic and make a fool of myself	2.42	0.82	0.30	-0.43
3. I felt that I had nothing to look forward to	1.94	0.79	0.71	0.34
4. I found it difficult to relax	2.03	0.76	0.61	0.38
5. I felt down-hearted and blue	2.12	0.7	0.60	0.73
6. I felt I was close to panic	1.97	0.77	0.57	0.10
7. I was unable to become enthusiastic about anything	1.88	0.71	0.69	0.77
8. I felt scared without any good reason	1.91	0.83	0.71	0.00

Skew skewness, *Kur* Kurtosis**Fig. 1** Scree plot for exploratory factor analysis

3.2 Factorial validity with exploratory factor analysis

A maximum likelihood exploratory factor analysis was conducted to examine the factorial validity of DASS-8. Given that the original DASS-8 [65] conceptualizes depression, anxiety, and stress as theoretically related constructs, Promax rotation was specified as an a priori analytic choice. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy yielded a value of 0.88, and Bartlett's Test of sphericity produced a test value of 1923.43, $p < .001$, indicating that the current dataset was suitable for EFA.

The eigenvalues and scree plot (see Fig. 1) suggest a unidimensional structure for DASS-8, explaining 50.42% of the total variance. Parallel analysis further supported this

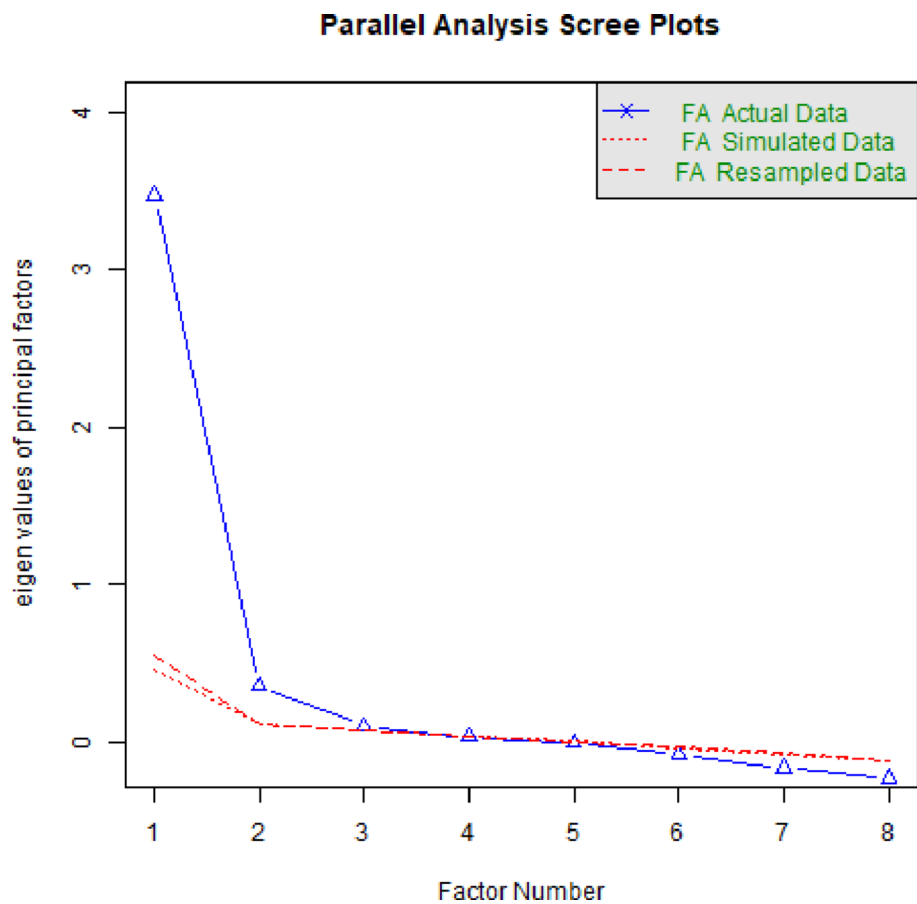


Fig. 2 Parallel analysis for DASS-8

Table 2 Exploratory factor analysis for DASS8

Item	Factor loading	Communalities
Item 1	0.61	0.37
Item 2	0.60	0.36
Item 3	0.59	0.35
Item 4	0.69	0.47
Item 5	0.71	0.51
Item 6	0.72	0.52
Item 7	0.63	0.40
Item 8	0.70	0.49
Percentage of variance explained	50.42	
Eigenvalue	4.03	

solution, as only the first eigenvalue from the actual data exceeded the corresponding eigenvalues derived from both simulated and resampled datasets (see Fig. 2), while subsequent eigenvalues fell below those expected by chance.

Item communalities ranged from 0.35 to 0.52, indicating that most items were reasonably represented by the extracted factor. Generally, the items loaded well onto the uni-dimensional structure, with factor loadings ranging between 0.59 and 0.72. The DASS-8 exhibited strong internal consistency, as evidenced by a Cronbach’s alpha value of 0.86 (Table 2).

4 Study 2 methods

4.1 Participants and procedures

A separate sample comprising 202 participants took part in Study 2, with 56 males (28.87%), 146 females (75.26%). The inclusion criteria were that participants were Malaysian adults aged 18 or older, with no specific exclusion criteria. An opportunity sample of Malaysians above 18 years old ($M = 25.50$, $SD = 8.78$) was recruited through social media and university information boards. Participants voluntarily agreed to participate in an online survey after providing their informed consent. The online survey via SurveyMonkey was completed in the lab before (pre-test) and after (post-test) an experimental session in approximately 30 min. The study was approved by the university's research ethics committee. Data collection for Study 2 was conducted from November 2022 to February 2023.

4.2 Measurements

The same as in Study 1.

4.3 Statistical procedures

Descriptive analysis was performed using R 4.5.1 to examine mean, standard deviation, and data normality. Next, confirmatory factor analysis (CFA) was performed using *lavaan* and *semPlot* package to examine the internal structure of DASS-8. The factor structure was evaluated using a set of fit indices recommended by Jackson et al. [78], namely non-normed fit index (TLI), comparative fit index (CFI), root mean square error of estimation (RMSEA), and relative chi-square (χ^2/df). The cut-off point for good model was set as 0.9 and higher for TLI and CFI, 0.08 and below for RMSEA, while 5 and below for χ^2/df . Modification indices were examined for model improvement, with MI values greater than 3.84 used as the criterion for potential model modification. Residual covariances were added sequentially and re-evaluated after each step to minimize model overfitting. Cronbach's alpha was implemented to assess the internal consistency of DASS-8. Interclass correlation estimate was examined to evaluate short-term temporal stability of the DASS-8 over a 30-minute interval.

5 Study 2 results

5.1 Descriptive statistics

Table 3 displays the item-level descriptive statistics for DASS-8 using data from the pre-test sample. It is observed that all items displayed a normal distribution, supported by skewness and kurtosis values that fell within the acceptable range of ± 2 .

Table 3 Descriptive analysis for items in DASS8

Item	Mean	SD	Skew	Kur
1. I felt that I was using a lot of nervous energy	2.17	0.62	0.36	0.48
2. I was worried about situations in which I might panic and make a fool of myself	2.2	0.85	0.38	-0.44
3. I felt that I had nothing to look forward to	1.65	0.73	1.02	0.93
4. I found it difficult to relax	1.97	0.75	0.69	0.62
5. I felt down-hearted and blue	1.87	0.75	0.71	0.46
6. I felt I was close to panic	1.58	0.68	0.92	0.39
7. I was unable to become enthusiastic about anything	1.61	0.71	0.97	0.55
8. I felt scared without any good reason	1.57	0.69	0.96	0.38

Skew skewness, Kur Kurtosis

Table 4 Confirmatory factor analysis for DASS8

Item	Standardized factor loading	
	Model 1	Model 2
Item 1	0.613 [0.490, 0.712]	0.601 [0.483, 0.705]
Item 2	0.593 [0.474, 0.699]	0.583 [0.471, 0.684]
Item 3	0.623 [0.489, 0.740]	0.552 [0.429, 0.659]
Item 4	0.641 [0.510, 0.748]	0.653 [0.537, 0.752]
Item 5	0.752 [0.658, 0.824]	0.778 [0.686, 0.850]
Item 6	0.641 [0.503, 0.750]	0.671 [0.560, 0.764]
Item 7	0.652 [0.537, 0.758]	0.594 [0.483, 0.693]
Item 8	0.504 [0.355, 0.632]	0.567 [0.426, 0.686]
TLI	0.841	0.947
CFI	0.887	0.966
RMSEA	0.119	0.069
χ^2/df	3.85	1.96

TLI Tucker–Lewis Index, CFI Comparative Fit Index, RMSEA Root Mean Square Error of Approximation, χ^2/df = Relative chi-square

Table 5 Inter-class correlation using single rating, consistency, two-way mixed model

	Interclass correlation	95% Confidence Interval		F Test with True Value 0			
		Lower bound	Upper bound	Value	df1	df2	Sig
Single measures	0.87	0.83	0.90	14.34	201	201	0.000

5.2 Factorial validity with confirmatory factor analysis

CFA was conducted to assess the single-factor structure of DASS-8 as derived from Study 1. The fit indices for the single-factor structure generally fell below the predetermined cutoff points (refer Table 4). Upon reviewing the modification indices, potential relationships between residuals were noted, specifically between Item 3 (“I felt that I had nothing to look forward to”) and Item 7 (“I was unable to become enthusiastic about anything”), as well as between Item 5 (“I felt down-hearted and blue”) and Item 8 (“I felt scared without any good reason”).

To address these potential relationships, Model 2 was developed with additional residual covariances between the specified items. Despite the borderline RMSEA, Model 2 demonstrated satisfactory fit indices. All items loaded well onto the single-factor structure, with standardized factor loadings ranging between 0.552 and 0.778.

Furthermore, bootstrapping with 5,000 bootstrap samples was employed to assess the stability of the single-factor DASS-8. The absence of zeros within the 95% bias-corrected confidence intervals provided evidence for the stability of the single-factor structure.

5.3 Reliability

The DASS-8 demonstrates robust internal consistency, evident at both the pre-test (Cronbach’s alpha = 0.84) and post-test (Cronbach’s alpha = 0.85) stages. A significant positive correlation was observed between pre-test and post-test DASS-8 scores, with $r(200) = 0.87$, $p < .001$. Furthermore, an interclass correlation estimate was computed using a single rating, consistency, two-way mixed model. Referring to Table 5, the significant interclass correlation supports the retest reliability of the DASS-8.

6 Discussion

The current study aimed to examine the psychometric properties of the DASS-8 [65] in a Malaysian sample. Exploratory factor analyses, confirmatory factor analyses and reliability analyses (Cronbach's alpha, short-term temporal reliability, and interclass correlation) were conducted to evaluate the psychometric properties of the scales.

Interestingly, the current findings suggest that DASS-8 has a unidimensional structure factor. While the three-factor structure (depression, anxiety, stress) was observed in an earlier study [65], the EFA and CFA in the current study provides preliminary support for a unidimensional structure. The current findings suggest that the three-factor DASS-8 [65] may not generalise to Malaysian samples, where depression, anxiety and stress may be conceptualised and manifested differently. Taken as a whole, these findings suggest that the DASS-8 might be useful as a brief measure of general psychological distress, rather than as a tool for distinguishing between depression, anxiety, and stress as separate constructs. Nevertheless, future research should empirically examine this explanation further.

Indeed, the findings from Study 1 and 2 are consistent with several previous studies which have suggested that shortened DASS versions may be effective rapid-screening tools to measure a general anxiety-depression factor in Asian populations [76, 77].

The unidimensional structure in the current study contrasts with longer DASS versions, specifically DASS-21 and DASS-42, that have sometimes yielded two dimensions in Asian samples [51, 52], which in turn load on a higher-order anxiety-depression or general distress factor. Given that Asian populations may be less emotionally expressive than Western populations [22], it may be possible that the unidimensionality of DASS-8 in the current study reflects a cultural attenuation of the distinctions made between the symptoms associated with depression, anxiety and stress in a Malaysian sample. Alternatively, it may also be possible that DASS-8 does not contain items that sufficiently differentiate between depression, anxiety and stress, yet that are able to tap generalised emotional distress or negative affect. The fact that Malay translations and validations of the DASS-21 have yielded good Cronbach alphas for three factors [79, 80, 81, 82] suggests that this interpretation may be correct. Further studies comparing psychometric properties of translations of the DASS-21 and DASS-8 are warranted.

6.1 Strengths and implications

To our knowledge, the current study is the first validation of the relatively-recently-developed DASS-8 in a Malaysian sample. Notably, the unidimensional structure may reflect cross-cultural differences in how psychological distress is experienced and expressed, particularly in Southeast Asian contexts where emotional distress may be understood more holistically rather than as separate, distinct categories.

Rather than assuming the three-factor structure proposed by the original DASS-8 [65], researchers and clinicians should consider population-specific validation studies. The unidimensional structure observed in our Malaysian sample suggests that the DASS-8 may be most appropriately used as a measure of general psychological distress rather than a tool for differentiating between depression, anxiety, and stress in this population.

The robust psychometric characteristics of the DASS-8 in this non-clinical sample suggests that it could potentially provide a quick, reliable way to screen participants in those situations when it is not essential to separate anxiety, depression, and stress into

distinct categories. This may make it especially useful for studies where time is especially limited, or when researchers are more interested in identifying or screening for general psychological distress rather than pinpointing specific types of symptoms.

While further validation in clinical populations is warranted, the DASS-8 may show potential as a screening tool for healthcare providers and educators, particularly for online screening. In non-clinical or community settings, this type of brief assessment could be helpful for remotely assessing people who may already be experiencing psychological distress, and as a means to reduce barriers to help-seeking. A potential application could involve those with high DASS-8 scores being referred for more detailed mental health screening, creating a simple two-step screening process with minimal effort from the patient or participant compared to longer screening questionnaires. However, future research should establish appropriate cut-off scores and validate the DASS-8 in clinical samples before implementing such screening protocols.

6.2 Limitations and future directions

While the DASS-8 showed strong psychometric properties as a potential standalone, unidimensional scale for psychological distress, some limitations should be acknowledged when interpreting the findings. While comprehensive psychometric analyses including short-term temporal reliability and interclass correlation coefficients were conducted, future studies should examine the extent to which a single factor displays convergent or discriminant validity through correlations with other validated questionnaires of related constructs. Furthermore, the confirmatory factor analysis in the current study did not include comparisons with alternative theoretical models, and thus caution should be applied when interpreting the findings.

The scope of our sample, while providing robust initial validation data, may not have captured the full diversity of Malaysian populations and may not generalise across all groups. While the use of an English-only instrument may limit generalisability to non-English speaking segments of the Malaysian population, the present study is, to our knowledge, the first study validating the DASS-8 in the Malaysian context and may be an important stepping stone for future validation studies on this brief scale. Thus, it would be helpful to validate the DASS-8 with different groups—including both clinical patients and healthy populations, different socioeconomic backgrounds, as well as people of various ages—to examine if the unidimensional structure is stable across different samples. Further, the present study assessed short-term temporal stability using a 30-minute interval within a single session. While the short interval allowed us to examine whether the DASS-8 produces consistent measurements without interference from life events or other time-related factors, less is known about its stability over longer periods of time. Future studies would benefit from examining test-retest reliability of the DASS-8 over a longer interval (e.g. a few weeks) to ensure that the scale remains consistent when assessing psychological distress over different time points.

Since the DASS-8 is a shortened version of the DASS-21 and that validated Malay versions of the DASS-21 with three factors already exists [68, 60], there is also a possibility that the reduced number of items may have reduced the effectiveness of the DASS-8 to differentiate between depression, anxiety and stress as separate constructs in a Malaysian sample. Future research should translate the DASS-8 into Malay and directly compare its psychometric properties with those of the established Malay DASS-21. This

would help to determine whether the shortened version maintains its reliability and validity when adapted for local language use, and whether the brief format works as well culturally as the longer established version.

7 Conclusion

In conclusion, the current study provides the first validation of the DASS-8 [65] in a Malaysian population. While the unidimensional factor structure in this study deviates from longer versions (DASS-21 and DASS-42) that usually show three distinct factors, our findings suggest that the DASS-8 may instead measure general psychological distress. The DASS-8's brevity and strong psychometric properties as a unidimensional scale for psychological distress may make it a valuable screening tool in healthcare and research settings, as its unidimensional structure would simplify interpretation, reduce participants' mental fatigue, lend itself to online delivery to minimise stigma, and offer a quick, reliable tool for screening and initial assessment.

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Author contributions

EB, ASL, CMHDG conceptualised the studies and methodology. JYF analysed the data. EB, ASL, DHJL, and JYF were major contributors in writing the first draft of the manuscript. All authors reviewed and approved the final manuscript.

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Data availability

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The studies were approved by the Sunway University Research Ethics Committee (reference number: SUREC 2022/029) on 27 April 2022. All participants provided written informed consent prior to participation in the study. This research was conducted ethically in accordance with the Malaysian Code of Responsible Conduct in Research (2nd Edition) and the Declaration of Helsinki.

Competing interests

The authors declare no competing interests.

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